Data Association for Multiple Hypothesis Tracking on a Neutral-Atom Quantum Computer

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Abstract

Multiple Hypothesis Tracking (MHT) turns multi-object tracking into a combinatorial optimisation problem: choosing the best mutually compatible set of candidate tracks is a Maximum Weight Independent Set (MWIS) problem. MWIS maps naturally onto neutral-atom hardware, because the Rydberg blockade physically forbids two nearby atoms from both being excited. We implement the full hybrid pipeline — detection, gating, track scoring, clustering, geometric embedding, and an adiabatic sweep on a simulated Pasqal device — and test it on synthetic radar data and on real microscopy data from the Cell Tracking Challenge. The optimisation works: on every cluster small enough to emulate, the atom array returned the exact optimum. The proposed advantage did not materialise. In a controlled experiment measuring how quickly each method discovers the posterior over global hypotheses, the array needed 27.5 samples on average against 23.6 for simulated annealing. We trace the bottleneck to a single cause: real MHT conflict graphs are not unit-disk graphs, so the blockade cannot enforce all constraints and 34% of measurements had to be repaired classically. We argue this makes native unit-disk problem formulation a precondition rather than an optimisation.

1 The problem

A sensor produces, at each time step, a set of unlabelled detections. Nothing indicates which object produced which detection, some objects are missed, and some detections are spurious. *Data association* is the problem of deciding which detections belong together.

Committing to an association early is dangerous: if two objects cross and their identities are swapped, the error propagates forever. MHT avoids this by keeping many competing explanations alive and deferring the decision. The cost is combinatorial growth, and that is where a quantum computer was proposed to help.

Vocabulary. A *track* is one hypothesised history of one object: a sequence of detections, at most one per scan. A *family* is all tracks descended from the same initiating detection, i.e. all competing stories about one putative object. A *global hypothesis* is a set of tracks that could all be true at once.

2 From tracking to a graph problem

Following Papageorgiou and Salpukas [1], we build a graph in which each vertex is a candidate track weighted by its score, and an edge joins two tracks that are incompatible — meaning they claim the same detection, since one detection cannot come from two objects. The best global hypothesis is then

$$\max \sum_i w_i x_i \quad \text{s.t.} \quad x_i + x_j \le 1 \ \ \forall (i,j) \in E, \tag{1}$$

with $x_i \in \{0,1\}$ indicating whether track i is selected. This is the Maximum Weight Independent Set problem.

The weight w_i is the track's log-likelihood ratio (LLR): how much more likely it is that this sequence of detections came from a real object than from independent false alarms. It is updated recursively,

$$\mathrm{LLR}(k) = \mathrm{LLR}(k-1) + \Delta\mathrm{LLR}(k), \tag{2}$$

$$\Delta\mathrm{LLR} = \begin{cases} \log(1 - P_D) & \text{no detection,} \\ \log \dfrac{P_D\, \mathcal{N}(\nu; 0, S)}{\beta_{\mathrm{FA}}} & \text{detection used,} \end{cases} \tag{3}$$

where P_D is the probability of detection, ν the innovation, S its covariance and β_{FA} the spatial density of false alarms. The false-alarm density sets the scale: in a clean environment a coincidence is meaningful, in a cluttered one it is not.

Two further quantities matter, because MHT wants more than the single best answer. The probability of a global hypothesis and of an individual track are

$$P\{H_j\} = \frac{e^{L_{H_j}}}{1 + \sum_i e^{L_{H_i}}}, \qquad P\{T_i\} = \sum_{j\,:\,T_i \in H_j} P\{H_j\}. \tag{4}$$

Computing these requires a *population* of hypotheses, not an `argmax`. This is the property the quantum approach was expected to exploit.

3 Why neutral atoms

Exciting an atom to a high principal quantum number gives it a large dipole moment. Two atoms closer than the *blockade radius*

$$R_b = \left(\frac{C_6}{\hbar\Omega}\right)^{1/6} \tag{5}$$

cannot both be excited: the doubly excited state is shifted out of resonance. Placing incompatible tracks within R_b of each other therefore enforces the constraint $x_i + x_j \leq 1$ in hardware, with no penalty term to tune. In the frozen limit the Hamiltonian is

$$H/\hbar = -\sum_i \delta_i n_i + \sum_{i<j} \frac{C_6}{r_{ij}^6} n_i n_j, \tag{6}$$

which is Eq. (1) with the penalty supplied by geometry.

Encoding the weights. Pasqal's Detuning Map Modulator (DMM) applies only *negative* local detuning, as a fixed spatial pattern $m_i \in [0,1]$ times a global waveform. Setting

$$m_i = 1 - \frac{w_i}{w_{\max}} \quad \Longrightarrow \quad \delta_i = \delta_{\max} \frac{w_i}{w_{\max}} \tag{7}$$

makes the detuning landscape proportional to the track scores. This requires $w_i > 0$, which the paper's rule of keeping only positive-score tracks supplies for free.

4 Pipeline

The implementation (`NielsBohrProject.py`) runs the full loop:

1. **Detection and gating.** Predict each track forward with a Kalman filter; reject detections outside the validation gate ($d^2 \leq 13.8$, χ^2 with 2 d.o.f.).
2. **Branching.** Extend each track with every gated detection, plus a missed-detection branch; every detection also starts a new family.
3. **Pruning.** Delete low-scoring tracks, cap branches per family, merge duplicates, and apply N-scan pruning.
4. **Clustering.** Split the conflict graph into connected components; each is an independent MWIS problem.
5. **Embedding.** Find 2-D atom coordinates realising the graph.
6. **Quantum solve.** Adiabatic sweep, measure, repeat.
7. **Feedback.** Use the resulting hypotheses for Eq. (4) and for pruning.

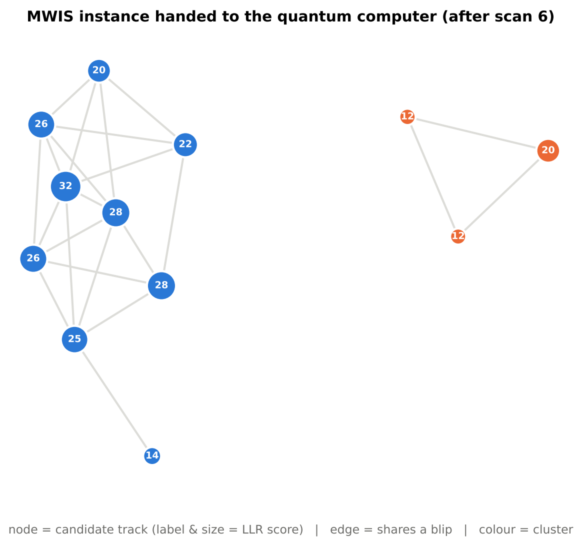

Figure 1: An MWIS instance produced by the tracker. Vertices are candidate tracks (label and size give the LLR score), edges mark tracks sharing a detection, colour marks the cluster.

Classical pruning is what makes it fit. The largest cluster over a whole scenario was 9 vertices, far inside the 256-atom device. Gating, confirmation, track merging and N-scan pruning — not the quantum step — are what keep the problem small. Removing track merging alone inflates clusters into near-cliques that no planar atom register can represent.

5 The geometric embedding

This is the hard part. On the hardware one does not choose the edges: physics decides them from distance, so the graph solved is always a *unit-disk graph*. An arbitrary MHT graph is not. We require

$$\|p_i - p_j\| < R_b \text{ (edge)}, \qquad \|p_i - p_j\| > R_b \text{ (non-edge)}, \tag{8}$$

and solve this by penalty minimisation with random restarts, working in units of R_b and choosing the physical scale afterwards.

That last point matters. The layout fixes only distance *ratios*; choosing R_b fixes everything else through Eq. (5). A larger R_b gives the atoms more room but costs Rabi frequency as R_b^{-6}, and too small an Ω never leaves the ground state within the device's 6 μs sequence limit. So Ω is *derived*, not chosen: we take the smallest R_b respecting the 5 μm tweezer spacing, then clamp it.

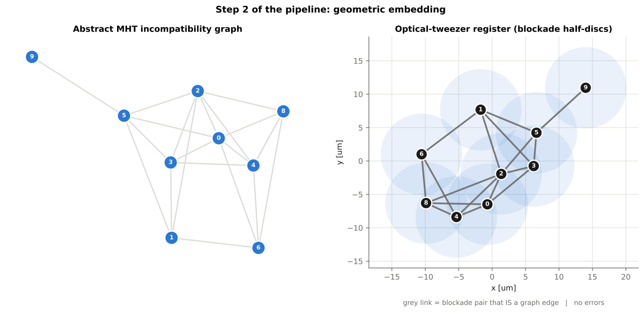

Figure 2: Abstract conflict graph (left) and the optical-tweezer register that realises it (right); shaded discs indicate the blockade range.

Cliques are the obstruction. A k-clique requires k atoms pairwise within R_b yet pairwise at least $d_{\min} = 5\,\mu$m apart. The feasible size depends only on $\rho = R_b/d_{\min} \approx 2.4$. A regular hexagon plus its centre gives seven points with ratio exactly 2, so at least seven fit; the true limit is around nine. Because branches of one family are mutually exclusive, each family contributes a clique — which is why capping branches per family is not a tuning choice but a hardware constraint.

Figure 3 shows how quickly perfect embeddings disappear.

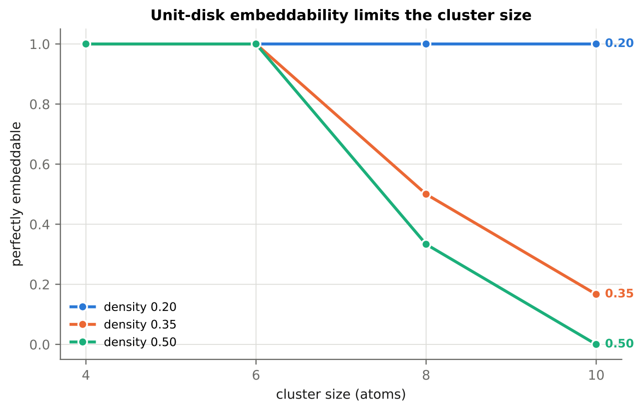

Figure 3: Fraction of random graphs admitting a perfect unit-disk embedding. The binding limit is cluster size and density, not the 256-atom capacity.

6 Validation

We reproduce the worked example of [1] (their Table 1 and Figure 5): six tracks with published incompatibility lists and scores, of which five have positive score. The published optimum is tracks $\{3, 6\}$ with weight 19.2. Our classical solver and the atom array both return exactly this, the array sampling it in 93–95% of shots with a defect-free embedding. This is the only external ground truth in the project.

In closed-loop tracking on synthetic radar data (five converging targets, $P_D = 0.9$, Poisson clutter), the array solved 21 MWIS instances and matched the exact classical optimum on all 21, tracking all objects with 100% association accuracy.

7 Testing the proposed advantage

The optimisation works, but at these sizes a classical solver finds the optimum instantly. The interesting claim concerns Eq. (4): the array should supply a *distribution* over hypotheses at fixed cost, whereas classical enumeration cost grows with how many hypotheses exist (up to $3^{n/3}$).

We tested this on real data: the `PhC-C2DL-PSC` sequence from the Cell Tracking Challenge [2], 300 frames of phase-contrast microscopy in which pancreatic stem cells proliferate from 74 to 661. Detections come from the raw images (background subtraction and connected components; recall ≈ 0.5, precision ≈ 0.85); the ground truth grades the detector only. From these we built six MWIS instances of 9–12 tracks.

Metric. We measure the *posterior mass* discovered per sample. Counting distinct hypotheses would treat a 0.1%-probability explanation as equal to one carrying 90%; Eq. (4) does not. A "sample" is one prepare–evolve–measure cycle for the array, one annealing run for simulated annealing, one construction for randomised greedy. All three receive identical post-processing.

Table 1: Samples needed to capture 90% of the posterior mass, averaged over six instances from real microscopy data.

Sampler	Samples	Precision
Simulated annealing	**23.6**	98.0%
Neutral atoms (repaired)	27.5	**97.8**%
Randomised greedy	35.3	91.0%
Neutral atoms (no repair)	32.2	64.2%

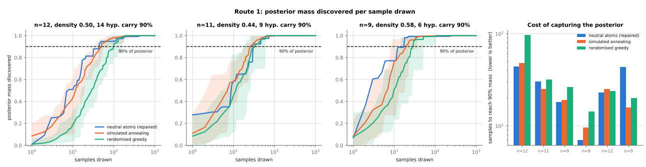

Figure 4: Posterior mass discovered against samples drawn.

Result. The claim is not supported. The array needed more samples than simulated annealing on average, though the margin is small and the reading should be "indistinguishable at this sample size" rather than a clear loss: the array won on four of six instances individually, beat randomised greedy comfortably, and annealing's lead vanishes when its budget is cut to two sweeps per sample. Six instances cannot settle the question either way.

8 Why, and what it implies

Embeddability dominates. The real conflict graphs had mean density 0.66, and 34% of raw measurements violated an edge. The blockade — the one mechanism meant to enforce constraints for free — was not enforcing them, because the layout could not realise every edge. Classical repair then does that work, and a repaired measurement is no longer a physical sample. This is exactly what Fig. 3 predicts.

Concentration fights coverage. Capturing 90% of the posterior needs about seven distinct hypotheses, but an adiabatic sweep is designed to return one. Figure 5 shows the trade: sweep duration moves $P(\text{optimum})$ from 8% to 99% while diversity moves the other way. The same register is an optimiser or a hypothesis generator depending only on how fast it is driven — a knob with no classical analogue, but not enough to close the gap here.

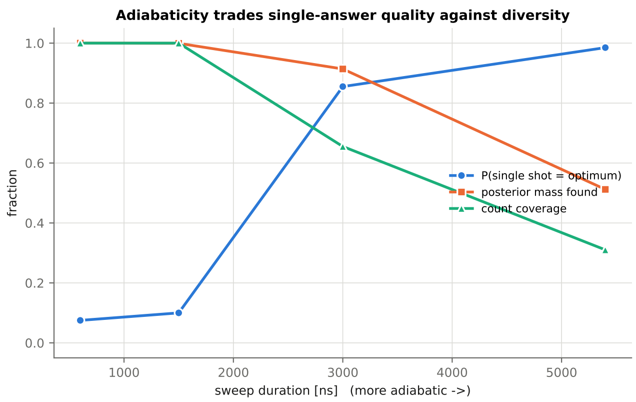

Figure 5: Adiabaticity trades single-shot quality against diversity.

λ is not a free parameter. The QUBO penalty is C_6/r_{ij}^6, fixed by atom placement. What one chooses is $\delta_{\max}/\Omega$, and it is bounded on both sides: too small and the detunings cannot express weight differences; too large and a blockaded pair profits from being excited together. The measured usable window was $2 \lesssim \delta_{\max}/\Omega \lesssim 5.6$, the upper bound being r_{edge}^{-6}.

Weight encoding is quantised. The DMM cannot resolve arbitrarily small local detunings, so score differences below about 2.6 LLR are invisible to the hardware.

9 Conclusions

The physics mapping is sound and the implementation reproduces published results exactly. What does not follow is an advantage. The limiting factor is not qubit count — we never needed more than 12 of 256 atoms — but geometry: MHT conflict graphs are unions of cliques joined sparsely, and cliques are the worst case for planar unit-disk embedding.

The useful consequence is that formulating the association problem so that the graph is *natively* unit-disk is a precondition, not a refinement. A promising and untested direction is to place atoms according to proximity in track state space, since tracks sharing a detection tend to be close there.

Open questions we did not resolve: the bias introduced by repairing measurements; unresolved targets, where two objects merging into one detection should be permitted to share it; and hardware noise, which attacks the weight encoding directly because the weights *are* detunings.

References

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